

Concorde Heritage Avionics Upgrade Package

(v3.00)



Manual for Concorde Heritage Avionics Upgrade Package v3.00,
Developer's Notes and Supplement to Manual
Colimata Concorde FXP 3.12.
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CONCORDE HERITAGE AVIONICS UPGRADE PACKAGE

Record of Revisions

Revision Number	Revision Date	Revised Section:
1	18.1.2023	All, initial
2	24.01.2023	All, corrections

Foreword

This manual includes procedures to operate some specific instrumentation that was either not documented in the original Concorde FXP package or was static equipment in the original package and is now no longer static but operational equipment under my Concorde Heritage Avionics Upgrade package.

When I first purchased the Colimata Concorde FXP version 1.00 several years ago I was disappointed with its lack of functions and the incomplete visuals. With my little knowledge of LUA programming, some basic 2D/3D modeling, along with help from community users, I was able to bring this much more fully featured Concorde aircraft to you. In its third iteration, this is the definitive Concorde sim flying experience now modified for Colimata Concorde v3.12 and X-Plane 12. As before (Heritage Avionics Upgrade Package v2.02 for X-Plane 11), I am offering it free, but donations are welcome. With this newest version a premium package is now also available.

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Chapters 1 - The List of Improvements Included in the Heritage Avionics Upgrade Package v3.00 over Colimata Concorde FXP v3.12

The following is a list of Concorde features that are now operational as a result of the Heritage Avionics Upgrade Package. These features were either entirely omitted from the original Colimata Concorde FXP v3.12 simulation or were present but received a significant functional upgrade and/or modification from the Heritage Avionic Upgrade Package.

- The Center of Gravity (CG%) computer is now fully functional and all associated displays and dials are operational
- Fuel Quantity Indicators for all fuel tanks are now operational
- Engine Control Schedule (ECS) now simulates and affects N₂ and Exhaust Gas Temperature (EGT) instruments
- Improved engine instruments behavior
- The Plessey PVS1580 Data Acquisition Computer is operational. This is the first ever implementation of this piece of Concorde equipment in any simulation.
- On the ground, visible windshield icing is now modeled
- All Three of Concorde's chronometers are capable of independent timing function
- The Automatic fuel trim transfer with load limiters is now functional
- Takeoff CG% 54% switch is now operational
- Improved Decision height alert algorithm functionality
- ISA (International Standard Atmosphere) temperature gauge is now also simulated and working on the Flight Engineer's (FE) panel
- Altitude, temperature, and speed display indicators are newly modeled on FE panel
- Improved exterior lightning and ambient cabin lightning
- Improved cabin temperature settings
- All gauges on the FE panel are now operational. Every system on the aircraft is now modelled but some fire, icing and undercarriage tests are not functional.
- As with the real Concorde, the aircraft's cabin pressurization system now requires flight crew involvement. You are now responsible for setting certain cabin pressurization parameters. (More details in another section.)
- The Oxygen system is one of four emergency systems onboard Concorde. It is now functional and O₂ is available from breathing masks in cases of emergency from hypoxia or blackout

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- Fuel management along with CG% fuel adjustments is now in your hands. Trimming the fuel by pumping is operational as is transferring fuel within tanks for landing.
- Refueling the system during a stopover is available and in same manner the jettison fuel system may also be used. (But I urge caution!)
- Automatic fuel trim settings are set by default, no need to go to main menu. Simply keep the CG% within the limits and it will be set automatically.
- Intake systems may be selected manually or automatically. However, in manual mode changing the ramps is still not possible.
- Engine status lights now react to overheating, engine failures, start and wind-down failures.
- Fuel temperature is now simulated. Temperature is regulated by the fuel heaters.
- Emergency re-light simulated for the relighting of engine failure when airborne.
- RAM air turbine and 5th emergency generator available in multi engine failure or power loss.
- Air conditioning system tests are now available and the results are indicated by corresponding warning lights on the FE panel.
- The Fuel panels many status lights e.g., overfill, low pressure, and low level, are now all functional.
- Hydraulic and electric panels are the key for sufficient power to all systems. Both now have proper logic algorithms with backup systems. (Be sure to keep them functioning correctly with correct the power outage and pressure during your flight.)
- The FE panel lights now have three (3) tests switches. You will find some that do not illuminate. These currently have no functionality. Coming in the future!).
- The Master warning sounds may now be muted with the correct switch on the back of the FE panel. The volume may also be adjusted.
- You may now use the loudspeaker switch to change between the default and the Heritage warning systems.
- Various overheat, fire, and smoke tests may now be conducted from FE panel.
- The Autopilot modes for Go-around and Back-beam are still present but they are now able to be used as helpful autopilot modes as documented in the Concorde Flying Manual.

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- There are Enhanced warning lights and sounds on flight deck
- Logic for various switches and scenarios has been added including warning lights with sounds making it easier to determine what is operational and what has diminished functionality on the Concorde.
- Concorde's actual starting procedure have been implemented. Ground power is an operational requirement for starting your engines.
- Concorde's braking system has remodeled. The brakes can overheat. Use the brake fans to cool the brakes and prevent tire blow outs in worst case scenarios. The associated warning lights work and are now functional.
- The Brake tests and associated temperature gauges are also functional.
- The default starting positions of some switches including the reheats have been changed. Fix for reheats has been removed.
- As a result of its much higher cruising altitude Concorde was originally fitted with a Geiger-Muller radiation tube type detector. The Heritage Avionic Upgrade Package includes a working radiation detector. I have simulated a radiation level by random number generator in nonlinear scale so the risk is very low, and you may not see a red warning ever. However, if you do see a warning, you need to act accordingly and descend to a lower altitude to reduce radiation exposure.
- Icing conditions warnings: A) As part of warning lights, the ice warning lights react to X-Plane default icing model. B) Because the default X-Plane icing model does not work while the aircraft is on ground, I recommend using my other addon. An Icing simulator that is more realistic and works well with the Concorde FXP. Now you can get visual icing on windows.
- Autopilot (AP) Enhancement:
 - o A) The Heritage Avionics Upgrade Package adds autopilot functionality for selecting the direction of the flight so you don't need to use your controller (joystick/VR) to turn the airplane left or right.
 - o B) It overrides the joystick turns if AP1 is ON. If AP1 is OFF, you are back to flying "hands-on" with your controller or joystick.
 - o C) Use black AP fine tuning knob (pilot's side, center console) turning the knob easily left or right and Concorde will start turning. There is a bank limit of 30 degrees.

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- Separate Power to Engines: An additional added feature is a separate file, “4heavy.lua”, which may be used with all aircrafts* but is especially handy with Concorde. It allows you to use only 2 of four engines on each side of the wing while turning. By default, if you have dual throttle controller all 4 engines are bound together, this will separate them into two pairs. 2 on the left and 2 on right. Compatibility with 2 or one engine powered planes is maintained. Extra/modified profile in X-Plane 11 controller config needed. Instructions below. Does not work with single or quad throttle axis controllers and in VR, only for dual throttle. (*-used by profile for 4 engines planes, other planes unaffected)
- Correct Vertical Speed (V/S) selector:
 - o A) Another feature is the enhance vertical speed hold. Use it like in a real aircraft. Use the up/down selectors next to the Auto Pilot (AP) knob.
- Enhanced Heritage audio sounds and radio mode:
 - o A) Sound recording between the tower and the BAW Concorde during its last flight are playable thru the radio comms channel. Many of these recordings are from JFK to LHR are played automatically. Only clearance, takeoff and one special recording may be recalled whenever you wish.
 - o B) These historic ATC recordings bring more authenticity to this famous plane.
- For IVAO users you can now benefit from 8.33kHz model on COM1/COM2 with the patch of the radio, also the frequency range is now corrected from 118 to 137 MHz
- Exterior and Interior Lighting Enhancement:
 - o A) The outside lighting of the aircraft has been reworked to a more realistic standard. This adds a spectacular effect for landing and taxi lighting effects.
 - o B) The package also contains corrected strobe light illumination effects.
 - o C) The final lighting patch is corrected cockpit and FE panel colors on the warning and alert lights. The original release has them mostly red which is inaccurate.

Chapters 2 - Getting Started with the Heritage Avionics Upgrade Package

Section 1 - Powering up

Is this your first time in the Pilot's seat? Whether your answer is yes or no there are some good practices here for you to learn. Follow the checklist provided by the Colimata team for other steps, set the route plan and adjust the autopilot etc. The following steps are only specific procedures which differs or are not documented.

Before providing power to the aircraft, on the fuel panel check to see if the switches are at their off positions. Some fuel pumps could be in the "ON" position, turn them OFF to prevent unwanted operation. The same is true for valves in the OPEN or AUTO position, turn them to the OFF position also.



(Fig. 1)



(Fig. 2)

To save space in the aircraft Concorde was not fitted with an Auxiliary Power Unit (APU). Therefore, it must rely on external power to begin the power up process. And the first step in this process is to call for that external power. You need a ground cart or truck to provide the aircraft with 115V power. You call it by depressing the yellow GRND Call button (Fig. 1) on the forward overhead panel. Additionally, on the Flight Engineer's (FE) left leg panel next to Ram Air Turbine switches is the Microphone selector (Fig. 2), switch it to ground. Now you will see on the FE electrical panel (on the righthand side) that the GROUND POWER AVAILABLE light is illuminated (yellow).

Once this has occurred, turn ON both batteries, A and B, on FE electrical panel. And finally, there are two switches that must be activated on this panel. Both the "Ground Power" and the SSB switches must be set to "CLOSED". The SSB switch is just below the Ground Power switch (Fig. 3). Now, sufficient power is being supplied to the aircraft to continue the power up (startup) process.



(Fig. 3)

Section 2 - Setting the Center of Gravity (CG%) Computer

Before engine start, the Center of Gravity (CG%) computer (Fig. 4) should be set. In the center of FE panel is the CG% unit with computer and weight selectors.



(Fig. 4)



(Fig. 5)

One knob is labelled ZFW (Zero Fuel Weight). This is the loaded aircraft weight without fuel. If you don't know this value, you can get it from Plessey Data Computer (Fig. 5) which is located on FE Right Leg panel. Details in Plessey PVS1580 section. After you obtain the value, input this value by turning the knob above the label for ZFW. After you have completed the data input, push the button to lock in the value. ZFCG (Zero Fuel Center of Gravity) is similar but it is currently predefined inside of X-Plane and entering the value has no effect. A/C (Aircraft) weight needs to be set in the display (Fig 6). The knob is at the N mark. Turn it counterclockwise to each dot from N and set ZFW. At every dot push and hold for one second left hand knob until the number changes.



(Fig. 6)



(Fig. 7)

Be very careful here, because the ZFW is displaying 10x lower than real ZFW, for example 88330kg your entry is 08833. When you reach 5th and last dot, go back to 4th dot, this will be the starting point for setting FUEL REM display which is the actual fuel on board. Again, 10x less than original, so 94123kg is 9412. Again, push the left-hand button as before and at every remaining dot clockwise. When you get back to "N" the fuel will be added to the A/C weight (Fig 7). From this point forward any fuel burned will be deducted from these displays showing weight and fuel decreases.

If you make an entry error, you can push and hold the N left hand knob for than 3 seconds and it will erase all the readings you have entered and you may start again.

Section 3 - Automatic and Manual Fuel Trimming Systems

“Trim for Takeoff”

After you have started the engines, you will hear the call out "trim for takeoff" announced by “Virtual Flight Engineer”. If you do not hear this call out it means your CG% is in front (ahead or forward of the front CG% mark) and must be adjusted (aft or toward the rear).

You would not want to ask your passengers to take a different seat so it would be preferable for you to move fuel aft! To do this you need to operate some fuel pumps and tank inlet valves.

You can do this manually or you can use an automatic setting. To do this manually, open the inlet valves on Tank 11 (Fig. 8) and turn ON the pumps on Tank 9 (Fig. 9) if it has fuel. This will move the fuel aft. (Tank 11 is located in the tail of the aircraft and Tank 9 is located forward near where the forward edge of the Delta wing joins the fuselage.)



(Fig. 8)



(Fig. 9)

To use the automatic process to change (lower/raise) load limits and automatically adjust the trim, use the switch located under the red cover (covering this switch is referred to as guarded) and move it to the appropriate position for the fuel transfer in this case “Rearward”.



(Fig. 10)



(Fig. 11)

Colimata implemented for ease of use a Trim callouts/settings system for beginners. You can set it from the menu or leave it in the automatic mode which is monitoring the current CG% position as fuel is moving and being used and other engine rating settings.

Here is the sequence and requirements as they are currently set within the system:

- Takeoff: CG% must be >52.7% and Engine rating switches at overhead panel set to Takeoff (i.e. Fig. 12)
- Flight: CG% must be >53.5% and Engine rating switches at overhead panel set to Flight, at or above FL 180 (if not serious stability issues will occur. They be overridden by menu preset) (i.e. Fig. 12)
- Descend: not defined (i.e. Fig. 12)
- Landing: CG% must be <55% and Engine rating switches at overhead panel set to Takeoff (i.e. Fig. 12)
- Ground: CG% <52.7% and ground speed <100 knots (i.e. Fig. 12)



(Fig. 12)



(Fig.13)

The automatic trim option only works if the switch on the “FUEL LP PROTECTION “ Bypass panel is placed in the “ARMED” position (Fig. 13) . With the automatic trim system deactivated, you will have to use Concorde menu presets for each trim state. Useful if not work for you.

The FUEL LP PROTECTION Bypass is also mandatory if you want to use the manual trim settings from menu. This is not recommended. The manual preset mode is not optimized in Heritage Avionics Upgrade Package. Many of the new features are disabled if you choose the Drop-Down Menus MANUAL mode or the Graphic User Interface (GUI) mode.

One final comment on Fuel Trimming but it is a very important one. Through the course of a normal Concorde flight, fuel is constantly being moved around to maintain the aircrafts proper CG% to optimize performance (See the % above). I strongly recommend you find a diagram of the Concorde’s fuel system with her pumps and inlet valves until you get used managing this system. It will greatly increase your learning curve and make flying this wonderful aircraft that much more fun.

Section 4 - Overlooked “Before Takeoff” Procedures

Before takeoff there are some important procedures which you will not find in standard Colimata FXP.

- Cabin Pressurization Control System

Two critical presets that are a part of the Cabin Pressure Control System are the Cabin Pressure Altitude and Cabin Rate of Change. You can set these after takeoff but there is too much to be done after takeoff and during your initial climb phase of your flight. It is easily forgotten. I recommend that you that you take care of these setting before takeoff.

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Top left side of FE panel there are System 1 and 2 gauges. Use the one of your choice to select desired cabin altitude for the pressurization of the system. Using Altitude Selector “A” sets the required altitude, usually 6.000 ft. Altitude Selector “R” sets the rate of change of cabin altitude, the white dot or 180° turn clockwise is about 400 feet per minute (Fig.14). This is an acceptable setting. The indicator will change after takeoff and it takes some time to reach the preset. At about FL150 the cabin altitude change rate should automatically steady at 0.



(Fig.14)



(Fig.15)

Be aware, if you reach a “cabin altitude” of 10,000 feet or above! You will receive a Master Warning Light, “EXCESS ALT LIGHT” (Red) and an audio (gong). This is also accompanied by another distinctive audio (intermittent horn). With the Heritage Avionics Upgrade Package, if the cabin is not fully pressurized you will find yourself in an emergency situation. This will require the use of one of the three (3) oxygen masks located in the cockpit. However, before you can use them, you must activate the emergency system. You do this by turning the black rotary selector located at bottom right of the FE panel (Fig 15). It must be turned to the crew’s position and the oxygen pressure must read about 70 PSI on the gauge. If this emergency is not immediately addressed and you do not put on an oxygen mask, your screen will go black..... you have just “Blacked Out!”



(Fig. 16)



(Fig. 17)

- Takeoff Monitoring (T/O) Settings

The second procedure that tends to be overlooked involved the setting of the Takeoff monitor. To ensure all engines provide enough thrust for takeoff the calculated fuel flow and pressure must first be set into the system. There is a row of lights across the top of each N₂ gauge in the fuel panel. These are referred to as “Power Management Lights” The first light on the left of each row is the “Clear to go Light”. It illuminates green when all power requirements for takeoff have been met. Otherwise, the Clear to go Lights on primary engine instruments will not light up and takeoff is not recommended under these conditions.

To set minimal fuel flow, look for a small rotating knob located next to the fuel gauge for each engine (Fig 16). Simply turn it to desired value. There are currently two static preset minimal values; takeoffs with reheat approx. 22tons/h and takeoff without reheat approx.11tons/h.

The P7 or pressure preset values are entered on the FE panel. This panel is located on the left side with the secondary engine instruments (Fig. 17). Again, this is a static presets value which for a takeoff with reheats should be set at approx. 39 PSI and without reheats should be set at approx. 16 PSI. (maybe a bit more). After these weights (tons/hr.) and pressures (psi) values have been entered for all of the engines you can push (*Push to Arm, Pull to Inhibit*) the “Take-Off Monitor Control Button” and if all other parameters are correct the “Clear to go Light” for each engine will illuminate green on primary engine instruments.

Section 5 - Fuel Balancing During Climb and Cruise

In Section 3. We discussed the requirements of maintaining certain limits of CG% percentages to prevent instability problems. To achieve these CG% parameter changes fuel transfers were necessary. We simply moved fuel aft from tanks 9 and 10 and transferred fuel from them into tanks 5 and 7 or tank 11. We also can use tanks 5 and 7 to refill constantly tanks 1,2,3 and 4 which are the primary “feed tanks” (also called Collector Tanks) connected to their respective engines. This technique is the same as in original Concorde FXP. What is different is the aft fuel trim with load limiters presets in the center of the FE fuel panel. When this switch is unguarded (uncovered) it can be used to automatically move fuel “Forward” or “Rearward” (Fig. 18). If the load limiter parameters (the counters) are set to certain values, they will leave those preset values on the respected tanks to maintain future trimming. This automatic trimming is only possible between tanks 9 and 11 with very limited transfer to other tanks. If for some reason tanks 9, 10, and 11 should all be empty than your last chance for any fore to aft or reverse trim is to shift fuel between tanks 1 and 4. This will be possible if tanks 1 and 4 are not being automatically refilled and the level goes below 1.7 tons of fuel. As soon as they are refilled, if you maintain this level, the results will produce a small aft CG % move.



(Fig. 18)



(Fig. 19)

The Standby Inlet Valve switch serve a special purpose within the Heritage Avionics Upgrade Package (Fig. 19). They function as a refueling “signal” while Concorde is on the ground. (More on the topic of refueling later in this manual.)

In the Original Colimata FXP they don't have a refueling function and they are used for unusual fuel transfers between all tanks. You should always ensure tanks 1,2,3 and 4, the collector (feed) tanks are filled. If one of these tanks completely empties, their associated engine will shut off.

Please note several pumps have an "AUTO" setting. This is to facilitate the auto trim function of the "Trim Trans Auto Master" switch. If you are not intending to use this functionality you should switch these off. In the "AUTO" position it will function as if it were in an "ON" position. The same is true for inlet valves with "AUTO" settings.

Section 6 - Engine Control Schedule (ECS) and International Standard Atmosphere (ISA) temperatures

ECS (Fig. 20) monitors and regulates N_1 (the rotational speed of the low-pressure turbine and compressor spool expressed as a percentage of maximum normal operating RPM), N_2 (the rotational speed of the high-pressure turbine and compressor spool expressed as a percentage of maximum normal operating RPM), and EGT (Exhaust Gas Temperature) parameters of engines during various phases of flight for fuel efficiency and noise limits.



(Fig. 20)



(Fig.21)

Of key importance is to note that when the Concorde exceeds Mach 1.0 do not leave the ECS in the "Flyover" setting. Keep in mind that the Engine instruments are now reacting to Outside Air Temperature (OAT) and ISA temperatures (Fig. 21). In extreme conditions you may have engine instruments warnings and find that the throttles must be adjusted to meet safety limits. If they are exceeded, they will be logged into PVS1580 data recorder for further evaluation.

Chapters 3 - Additional Functioning Instrumentation in the Cockpit on the Pilot and Copilot Panels

Here I will cover some of the differences you will find in the Heritage Avionics Upgrade Package that are not include in Colimata Concorde FXP v 3.12 for the Pilot and Copilot cockpit flight panel areas.

Section 1 - Audio Warnings and Heritage Sounds

Many of the pilot's and copilot's audio knobs and switches have a number of different callouts from Concorde's last flight as a tribute to a great airplane. These provide a little fresh ambient experience when the flight becomes a little quiet and boring. By default, the sounds are at low volume. To make all of the audio recordings available turn volumes up on center console (Fig. 22). Some sounds will come up automatically during flight. Some are triggered with green push buttons or "TO" monitor switch during taxi. If you want more audio warnings look for loudspeaker switch at pilots left arm.



(Fig. 22)

Section 2 - Weather Radar

In my last Heritage package, I made weather radar available. It also had a ground proximity system. With this newest version of the Heritage Avionics Upgrade Package I have now added a warning system.

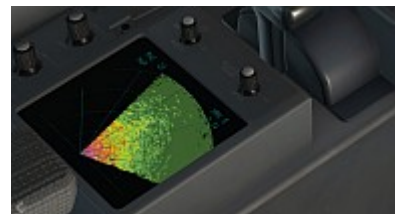
Only one function is available but showing same data on both units simultaneously. Turn the units on with the ON/OFF/TEST switch (Fig. 23). You can change the range on Pilots unit between 10 and 40 miles. To increase the viewing range use Copilot's left knob. With the Pilot's range set at 40, you can adjust the Copilot's range to reach out to 160 miles (Fig. 25). For weather display shows WXR, or WXA where "A" means visual alert of bad weather.



(Fig. 23)



(Fig. 24)



(Fig. 25)

Another alert is also available. Located to the copilot's right arm is the Radar alert switch. When activated, if bad weather is in vicinity a white flashing light appears near autopilot area. To have Ground Proximity Radar displayed on the screen (Fig 25), look on the center console where the other WX radar switches are located (Fig. 24). Switch SYS1 is WX radar, SYS2 is the Ground Proximity Radar. You may also adjust the antenna angle up or down and also the antenna gain. The switch to place the radars in standby or to turn them off are also located on this console. The radars need either ground power or the engines running to be operational.

Section 3 - Distance-To-Go (DTG), Counters and Timer

On center console is a green LED area used for Passenger information that is displayed in the cabin area for the passengers (Fig.26). It is not actually connected to the displays in the cabin but you can use it for your purposes.

It is excellent for measuring the remaining distance to your destination. Turn the black knob to set the correct distance to go in miles divided by 10. However, if you set this display to 300 what will be displayed in the cabin is 3000 nm. (Notice there is a painted zero.) When you switch it on by the switch located to the left of the display after takeoff it will start decreasing by the miles flown. It can be reset only after it reaches zero and may be reset as many times as necessary until it shows correct value.



(Fig. 26)



(Fig. 27)



(Fig. 28)

Timers are a new capability now that all three are independent clocks and operational (Fig. 27). The third is located at top left of the FE panel (Fig. 28). Elapsed time setting is available only at Pilots seat. The Chronometer and the timer for all three are functional. To use the timer, move the main switch from chrono to timer, then move the top switch from hold to the left positions where first is seconds and last is minutes. If you wait at one of positions it will start increase time automatically. Back to run will start countdown. To reset it go left to reset.

Section 4 - Enhanced Autopilot Modes

A **Go-Around Mode** is now available in the Heritage Avionics Upgrade Package. It is activated when you are in the landing phase of your flight, on final approach, below 1000 ft and you need to abort the landing. Don't panic, just push the throttles to full open and the Go-Around Mode will be activated automatically. The function will acquire and maintain a pitch 15 degrees until the aircraft reaches 1000 ft. The function will then terminate automatically. You can now turn on the back beam mode if you were on localizer to follow it after go-around, the limit is 30 miles. Vertical Speed (VS) may now be set from the center console at the autopilot control (Fig, 29), pitch datum. Flight Director (FD)1 must first be on to see the selection in VS indicator. With AP1 on the VS autopilot modes during flight will follow preset vertical speed.



(Fig. 29)

Turning the knob on the center console may also be used for lateral navigation if no other mode is set. Use this black rotating knob to turn the aircraft left or right. The resulting maximum bank angle is 30 degrees. For some reason this functionality is disabled when MAX CLIMB and/or MAX CRUISE are enabled.

Section 5 - Improved Decision Height and Waypoint Change Warnings

There is only one small improvement here. If you preset during your landing a Decision Height (DH) altitude with the red knob, you will hear a sound warning you when the DH has been reached.

The Waypoint Change alert will display a flashing yellow warning light near the autopilot area when you are within one minute from your next waypoint.

Section 6 - Icing

We implemented an on the ground visual icing on the windshield, Visor, and the side windows (Fig. 30 & Fig. 31). You can now see the build-up of ice during the weathers and in colder climates.

Both Visor and windshield are modeled. To get rid of ice simply turn on windshield de-ice at overhead panel.



(Fig. 30)



(Fig. 31)



(Fig. 32)

Another ICE warning and de-ice system in the overhead panel are connected to X-Plane icing system. I strongly recommend use of my Icing Simulator v1.23 available on x-plane forum for better icing experience because default X-Plane model is very poor.

Section 7 - Air Data Computers (ADC)

On the aft center console are ADC 1 and ADC 2 computers (Fig. 33). Their function is to compare the readouts of the various sensors and instruments and if there is discrepancy to notify the pilot. Both have various TEST modes, the longest takes 35 seconds and every test is completed when the blue TEST light is on. Most of the tests are available only at ground.



(Fig. 33)

Chapters 4 - Flight Engineer's Additional Functioning Instrumentation and Gauges

The Flight Engineer's panel was heavily modified before Colimata Concorde FXP v3.12. My first Heritage Avionic Package (v2.02) made many new flight functions available. I have built upon that foundation and continued to make more flight functions available to provide the most "realistic" Concorde flying experience possible. Here are even more extra

Section 1 - Brakes Gauge

In the top left corner of the FE panel, you will see gauges that provide information about Concorde's braking system. (Fig. 34)



(Fig. 34)

The Pressure Gauge provides information about the amount of pressure applied to the brake pedal. The second gauge, the Temperature Gauge, display the highest actual brake temperature amongst the brake groups or if a “BRAKES TEMP” light is pressed, the temperature of the associated brake. A brake lamp will illuminate (RED) if the brake temperature exceeds 220°C. There is test button for gauge test. If the brake temperature exceeds more than 700°C you risk losing tires to puncture from heat. (Note: There is also a “WHEELS Over/HEAT” indicator on the Copilot’s side of the cockpit.)

Section 2 - Intake Pressure Error Gauges



(Fig. 35)

Below the brake temperature gauge on the FE top left panel are four indicator gauges. These gauges are the “Intake Pressure Ratio Error” gauges (Fig. 35). They show the intake status at aircraft speeds above M 1.3 following an intake control error (failure), the pointer shows the direction in which the throttle lever is to be moved to obtain intake - engine matching. If the pointer lies in the left amber band the throttle lever must be retarded. If the pointer lies in the right amber band the throttle lever must be advanced. Correct intake - engine match is indicated when the pointer is between the amber bands.

Section 3 - In Flight Engine Restart Procedure



(Fig. 36)



(Fig. 37)

If for some reason one or more of your engines fails during flight and there is no permanent damage, you may try to restart them. With the Heritage Avionics Upgrade the equipment necessary to perform this action is modelled and no longer static. Here is the procedure. First you need to start the emergency generator whose controls are located on right of the FE electricity panel (Fig. 36). Unguard the switch and move it to the “ON” position. This action will cause a blue light to illuminate on the panel.

Then using the knob labelled “EMERGENCY RELIGHT BUSBAR” (Fig. 37) which is located on the bottom left of the FE panel below and right of the Engine Debow switches. Rotate the knob from the off position to the number that corresponds to the engine you wish to restart. Then go thru the normal engine start procedure. After a successful restart return the “EMERGENCY RELIGHT BUSBAR” knob to the off position and secure the emergency generator.

Section 4 - Additional Pressurization Switches.

In Chapter II Section 4 we described the basic functions of pressurization system. There is additional information that we now need to discuss.

We have taken the appropriate steps to properly pressurize the cabin before takeoff. Now, before landing, must be depressurized the cabin. To do this we will reverse the pressurization process. Turn the Cabin Altitude Selector “A” back down to zero, completely to zero and Altitude Selector “R” rate knob counterclockwise to the limit or until the rate goes to the negative values (Fig. 38). Perform these actions only after passing below FL150.



(Fig. 38)

The “SYSTEM SELECT” switches maintain themselves in pairs. SYS 2 is the backup system for SYS 1. Discharge valves for the respective systems are usually in the “NORM” position but may be changed to closed. Ditching for “SYS 1” and “SYS 2” and “EMERGENCY DEPRESS” are for emergency situations and works. Ditching is for the prevention of flooding the interior when an emergency landing on water and “EMERGENCY DEPRESS” is used as a rapid 4000 ft per minute depressurization mechanism. “EXCESS ALT” and overpressure warnings are shown when the system operates beyond limits.

Section 5 - Additional Information on the Engine Control Schedule (ECS)



(Fig.39)

There are four levels of performance indicated by the four levels of lights. These levels of performance are (Fig. 39):

- Low (LO) - is required during dry engine operation at speeds less than 220 knots and during reheated engine operation in T/OFF rating-at any speed; it also protects against excessive N_1 overspeed when reheat is cancelled. (*default maximum power, takeoffs, reheat climbs*)
- Mid - is required during reheated engine operation in CLIMB rating at speeds greater than 220 knots, Mid also protects against excessive N_1 overspeed when reheat is cancelled and during the approach for noise abatement to touch down. (*for landing approaches*)
- High (HI) - is required during dry engine operation at speeds greater than 220 knots. It provides maximum fuel economy. (*the opposite of LO by adjusting $N_1:N_2$ ratio for Max fuel efficiency during cruise*)
- Flyover (F/O) - is required during flyover noise abatement at speeds greater than 220 knots and for prolonged flying within the range M 0.80 to M 1.00 provided that the throttle levers are not advanced more than 95% of travel. (*for noise abatement procedures, for flying over land with <95% power*)

Section 6 - Fuel Tank Pressure Gauge and Radiation Gauge

On the FE left panel immediately below Engine Secondary Instruments panel you will find the Fuel Tank pressure gauge (Fig. 40). This gauge normally reads 0 but at or above FL400 the tank is automatically pressurized to about 1.5 PSI. If you climb too fast you can over pressurize the tank which will cause the Over Pressurization warning light to illuminate. In this case you must decrease your rate of climb to maintain 1.5 PSI. Same situation occurs during descent.



(Fig. 40)



(Fig.41)

This is in reality a randomly generated potential radiation emergency to simulate solar and ambient radiation. Radiation is higher at higher flight levels and closer to the poles. Concorde's home when not on the ground. Other commercial aircraft flying at lower altitudes would not have encountered this higher risk of radiation exposure as the Concorde's crew and passengers. Thus, Concorde was fitted with equipment (Fig. 41) to monitor and measure radiation exposure during her flights.

Section 7 - Refueling, CG% Computer Modes and Fuel Consumed Displays

Fuel should be loaded before powering up the aircraft. You can use Colimata GUI (Fig. 42) or simply default X-Plane weight and balance menu (Fig. 43) where you can only estimate the amount by percentage of the roller bar (because the units are wrong). There is the possibility of refueling after powering-up the aircraft but you have to apply a refueling procedure specific to the Heritage Avionics Upgrade Package. To do this you must; a) be parked on ground and b) place all the “STANDBY INLET VALVES” in middle of FE fuel panel to the “OPEN” position, all of them (Fig. 44). As the fuel in the tanks begins to fill, you will see that the total fuel contents counter differs between Captain’s/Copilot’s total fuel contents counter and the one on the FE panel. Normally, the refueling should stop when the limit is reached.



(Fig. 42)



(Fig. 43)



(Fig. 44)

If it does not, or your fuel requirement is less than a complete load out, you can stop it manually by closing that standby inlet valves. This procedure is used normally for refueling after stop overs.

On top of the CG% computer is a rotary selector knob (Fig.45) which permits numerous test to be selected by turning the knob on the computer. One such test is named Gauges. It will test the free movement of the needles. Next are several Min Max tests, these are only internal comparator tests, Fig. 45 specifically shows the comparator warnings generated by this test. And finally, 1 CG% and 2 CG% tests are again internal comparator tests.



(Fig. 45)

Filament test, “FIL CG%”, test the display elements and shows 88.8 at CG% display. After all tests have been completed, move the rotary knob to the “OFF” position and TEST switch to the “OFF”.

The CG% Channel Rotary Selector is normally at M position (Fig. 45). M selects the main computer for CG% system. Here is the information each CG% Channel provides:

- M – selects the main computer for CG% system. The main channel uses inputs of ZFW, ZFCG, and all (13) FQIs.
- 1 – selects standby computer 1 for the CG% system. The standby 1 channel uses inputs of ZFW, ZFCG and FQIs for tanks 1, 2, 5, 5A and 6 plus A channel in tanks 9, 10 and 11.
- 2 selects standby computer 2 for the CG% system. The standby 2 channel uses inputs of ZFW, ZFCG and FQIs for tanks 3, 4, 7, 7A and 8 plus B channel in tanks 9, 10 and 11.

If some of the gauges stop working or begin providing bad data, you may change the CG% channel. Set the channel to 1 if failed gauge is on the left side and select 2 if failed gauge is on right fuel panel. These modes 1 and 2 use mirrored copy of gauges. (These actions will not affect trim tanks 9,10 and 11).

(Note: FQI is Fuel Quantity Indication. The FQI system measures the fuel contents of the tanks by means of capacitance type gauging channels and provides individual indication of each tank content at the Flight Engineers fuel management panel or alternatively, at the re-fuel control panel for refueling.)

The Fuel Consumed displays (Fig 46) show the amount fuel burned by each engine. In addition to the display counter there is a knob used to test and reset the counter. To reset the Fuel Expended Counter, turn the knob to the left. It will reset to Zero. To test the display and the Fuel Density Compensation Error Indicators turn the knob to the right. Both the digits and the lights will illuminate. The digits will extinguish automatically after the test but you will have to turn the knob back to the left for the lights to extinguish.



(Fig. 46)

Fuel density is not simulated so it is completely random. Fuel “dump” or Jettison fuel can be done but is not recommended because Colimata put the discharge on a very high rate and you may lose too much fuel within a one minute.

Section 8 - Plessey PVS PVS1580 Data Acquisition Computer

This is the first ever simulation of this Air Data Computer System within any Concorde model. It is used for monitoring and diagnostics of performance data of engines, speed, etc., It uses 4 digits codes to query the performance data collected and stored during the flight. The list of the codes is accessed through the Main GUI menu under the custom content tab.

Concorde Heritage Avionics Upgrade Package

To use it on ground, as is usually done before takeoff or after landing, place the “Test Switch” on the bottom face of the panel to the left of the rotary dial to the “ON” position. *(Note: By default, the Plessey Data Acquisition Computer Test Switch will go to the “POWER” position after disconnecting from ground power. To see current performance data during the flight you will have to place the Test Switch back into the “ON” position.)*

Use keypad to enter the code representing the desired data, when all 4 digits have been entered, they will display in the left-most display character holders. Hit the “DIS” key located in the right most column and your data, if available, will be displayed in the right-most display character holders. If no data is available, four (4) zeros will be displayed in the right most display character holders.

To enter new code just start typing in the new code and the display will clear and begin entering the code. Then you may proceed as before. If you want to clear the data display, use “CLEAR” key. At the next Stop-Over or before the next flight use the “CLear MEMory” key to clear all stored and collected data from memory (doors must be open).



(Fig. 47)

The Rotary Fault Identifier Selector connects the corresponding unit with recording tapes to the Fault Key/Light (Amber) red fail light. SYS is general mode for complete system. However only AQ1 will not activate the Fault Key/Light because there are simply no more tapes available today and the old PDP 11 readers used for data collection are all in museums!

If you see a Green light (Limit Key/Light) that means some limits have been reached. On the code display, you may see which one or which category has reached its limit. If you depress the “DIS” on the keypad in this mode you will see the percentage of the limit that has been exceeded.



Section 9 - Emergency Oxygen System Gauges

At the bottom right of the FE panel near the floor there are pressure gauges for the passenger's and the flight deck area (Fig. 48). After takeoff, turn the black knob to open the oxygen system for the oxygen masks in the flight deck area in case of an emergency and to prevent the flight crew from blacking out. If you wear a mask for long time, the pressure will slowly decrease as the oxygen is consumed. The oxygen bottles only have a capacity of about 20 to 22 minutes. It can be refilled along with refueling procedure



(Fig,48)

Chapters 5 - Additional Features

Section 1 - ISA, International Standard Atmospheres

ISA (International Standard Atmosphere) temperature is normally not modeled in X-Plane. But because temperature is such a key performance indicator for Concorde's performance, particularly as she approaches and achieves speeds of Mach 2, it must be included. If the display shows ISA +09 (Fig. 49) it means the day is hotter than the standard and in hotter air Concorde's engines produce less power. For you, the pilot, this means you should carefully monitor engine temperature gauges and Total Temperature Indicator which is 127°C.

Be sure to conduct all of your warning lights tests, fire tests, air condition tests, smoke tests, etc., and others that are available. This should be done before every flight. And don't forget to turn the test selectors back to off after completing the test and beginning your flight.

Various tests of warning lights, fire tests, air condition tests, smoke tests are available, but not all are working (Fig. 50). This should be done before every flight. Don't forget to turn the selectors back to off before flight.



(Fig. 49)



(Fig. 50)



(Fig. 51)

Section 2 - Fuel Scavenging System

The fuel system has an automatic scavenger tank and pump system operating in the background, Scavenge Pump MI (Fig. 51). Notice the white “ON” Magnetic Indicator (MI) in the figure. This indicates the scavenger pump is running. The pump removes any fuel that has entered the scavenger tank from the main tanks, through the vent system, and returns it to Tank 3. Operation of the scavenger pump is automatic.

Chapters 6 - Notes

Section 1 - Failure Mode

X-Plane has a failure a mode available, this is set ON by default. You may with Heritage Avionics Upgrade Package occasionally see some systems failures because I made them to interconnect. And because X-Plane’s generated failures are usually too aggressive, I recommend in these cases you look for nearest airport and try to land. Or you may choose to simply disable the failure, fix them with one click, and continue. Some systems may fail even with the failures turned off because they may exceed run time limits.

I said that ECS and temperatures affect performance. But this is only true of the cockpit gauges. The real X-Plane model is untouched in this version. This means you still have 100% power by the X-Plane model even if the gauges are indicating to you that you are over or below limits. For now, this is for compatibility.

If you did not understand how all of the systems work, use checklists or watch my tutorial videos on YouTube.

Happy Flying!

Version 3.00

Gaston 2023

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CONCORDE HERITAGE AVIONICS PACKAGE UPGRADE

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